Chapter 11

A Real Intertemporal Model with Investment

Chapter 11 Topics

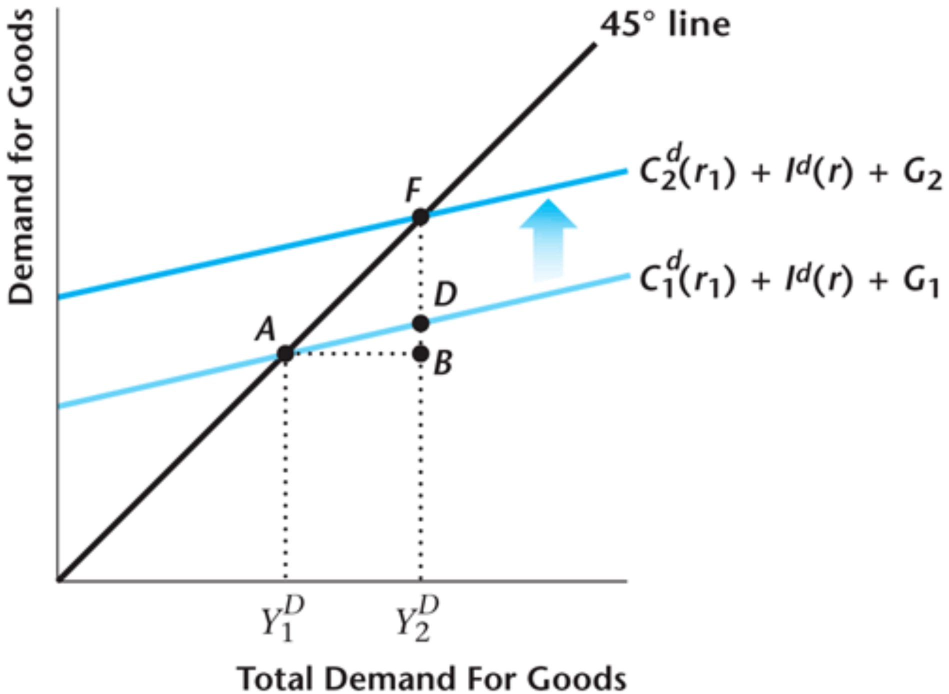
- Construct a real intertemporal model.
- Understand the investment decision of the firm.
- Show how macroeconomic shocks affect the economy.
- Focus on the implications of future expectations for current macroeconomic performance.

The Equilibrium Effects of a Temporary Increase in G

- Question: What is the rightward shift in the output demand curve due to an increase in G?
- Answer: The curve shifts to the right one-for-one with the increase in G.
- The *partial expenditure multiplier* is one.

$$\frac{Y_2^d - Y_1^d}{G_2 - G_1} = 1$$

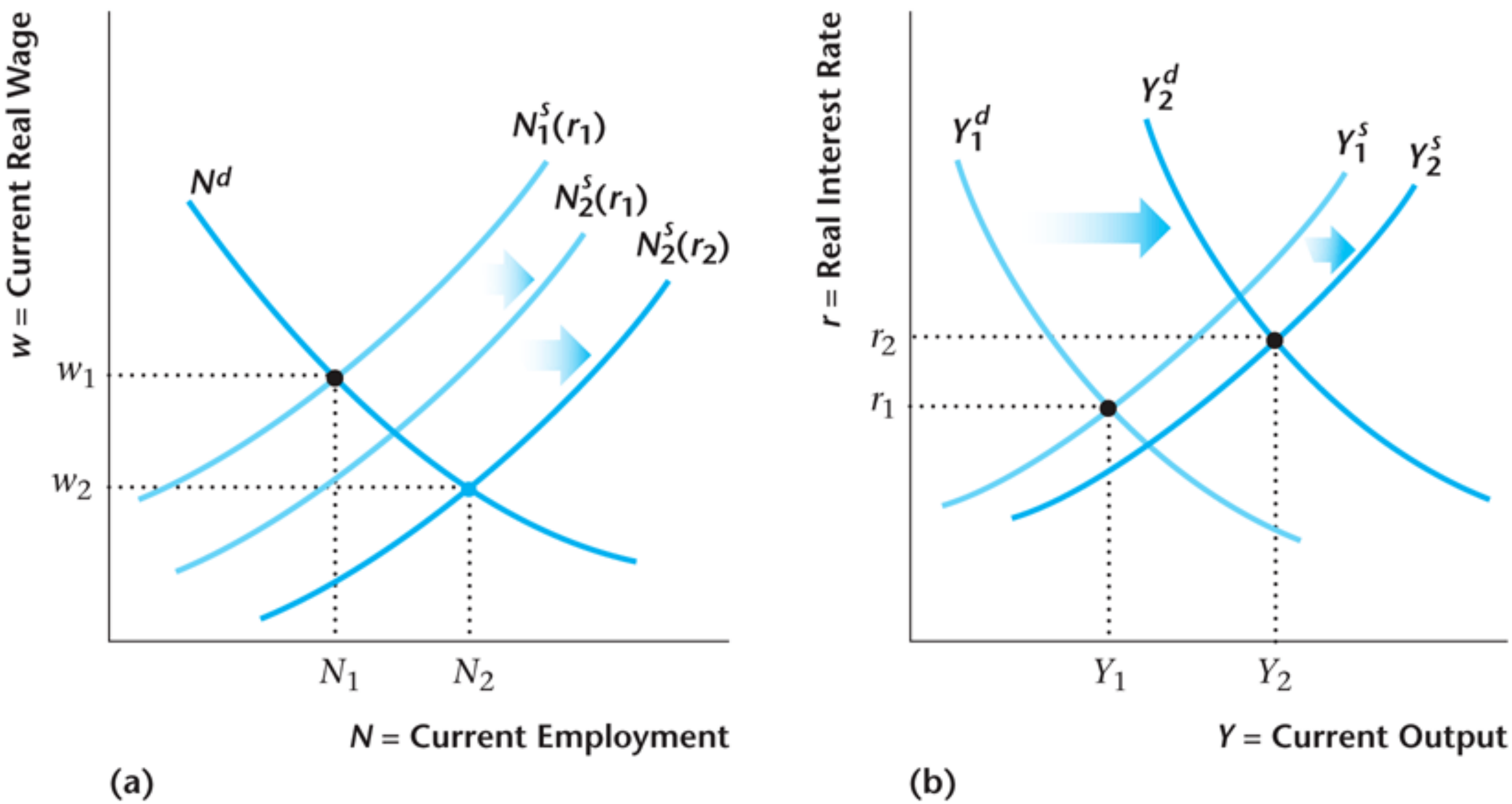
Figure 10.22
The Effect of an Increase in G on the Total Demand for Goods



Equilibrium Effects

- Output increases, real interest rate increases, real wage falls, consumption and investment decrease, employment rises.
- Government spending crowds out both consumption and investment.

Figure 10.23
A Temporary Increase in Government Purchases



Important Points

- The total multiplier – the ratio of the equilibrium increase in Y, to the increase in G – must be less than 1.
- In rudimentary Keynesian analysis, the multiplier is greater than 1. Why? That type of analysis does not take account of: (i) how the extra output is to be produced; (ii) the effect of taxes (present and future) on consumption.

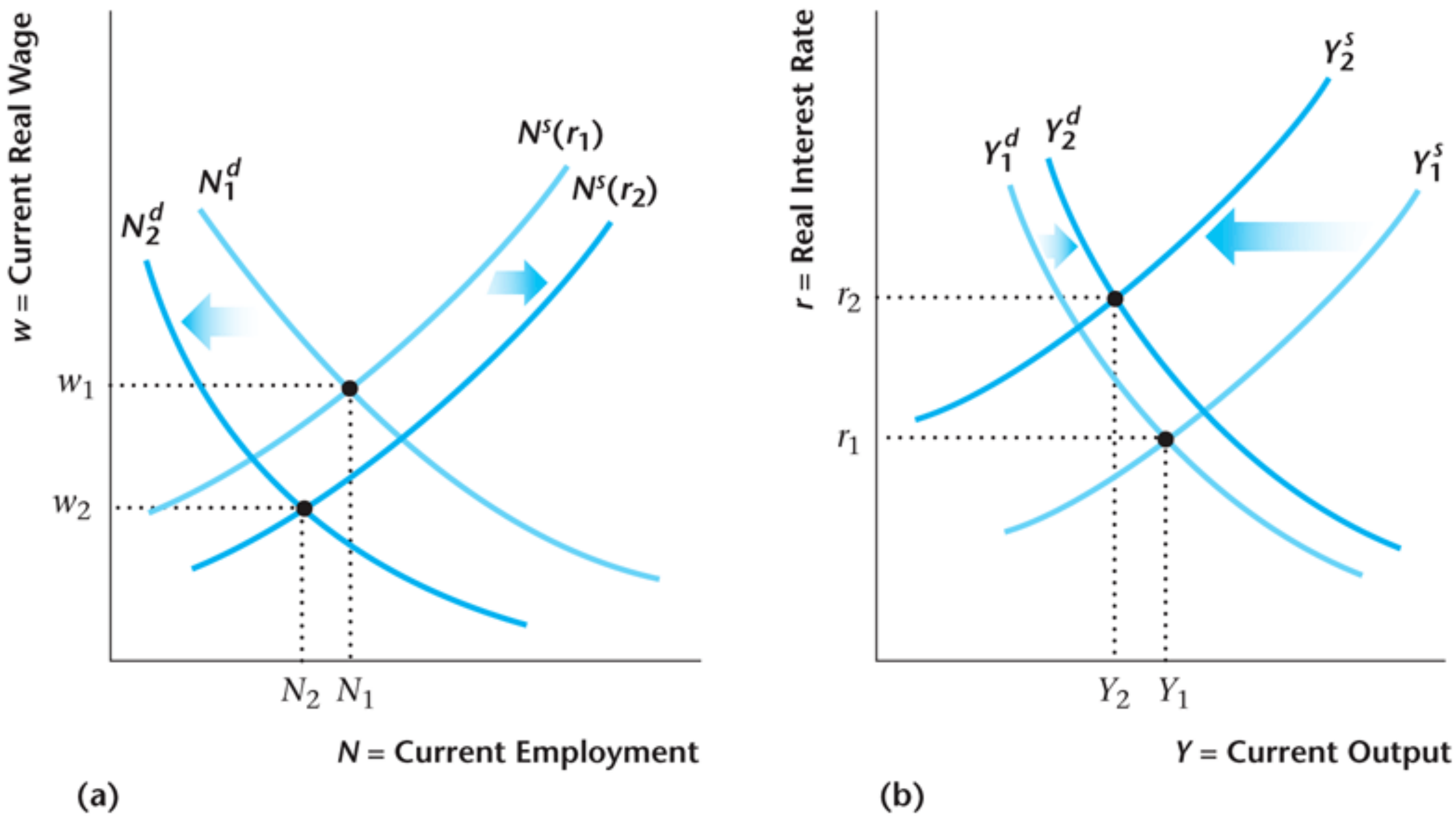
More Important Points

- Extra government spending comes at a cost – crowds out private consumption and investment spending, and we all have to work harder.
- In rudimentary Keynesian analysis, increases in G are costless – in fact Y and C increase as a result. The conclusion seems to be that G should be infinitely large – obviously silly.

A Decrease in the Current Capital Stock, K

- This could arise due to a war or natural disaster.
- Output may rise or fall, depending on how large the output demand effect is relative to the output supply effect.
- The real interest rate rises, the real wage falls, employment may rise or fall.

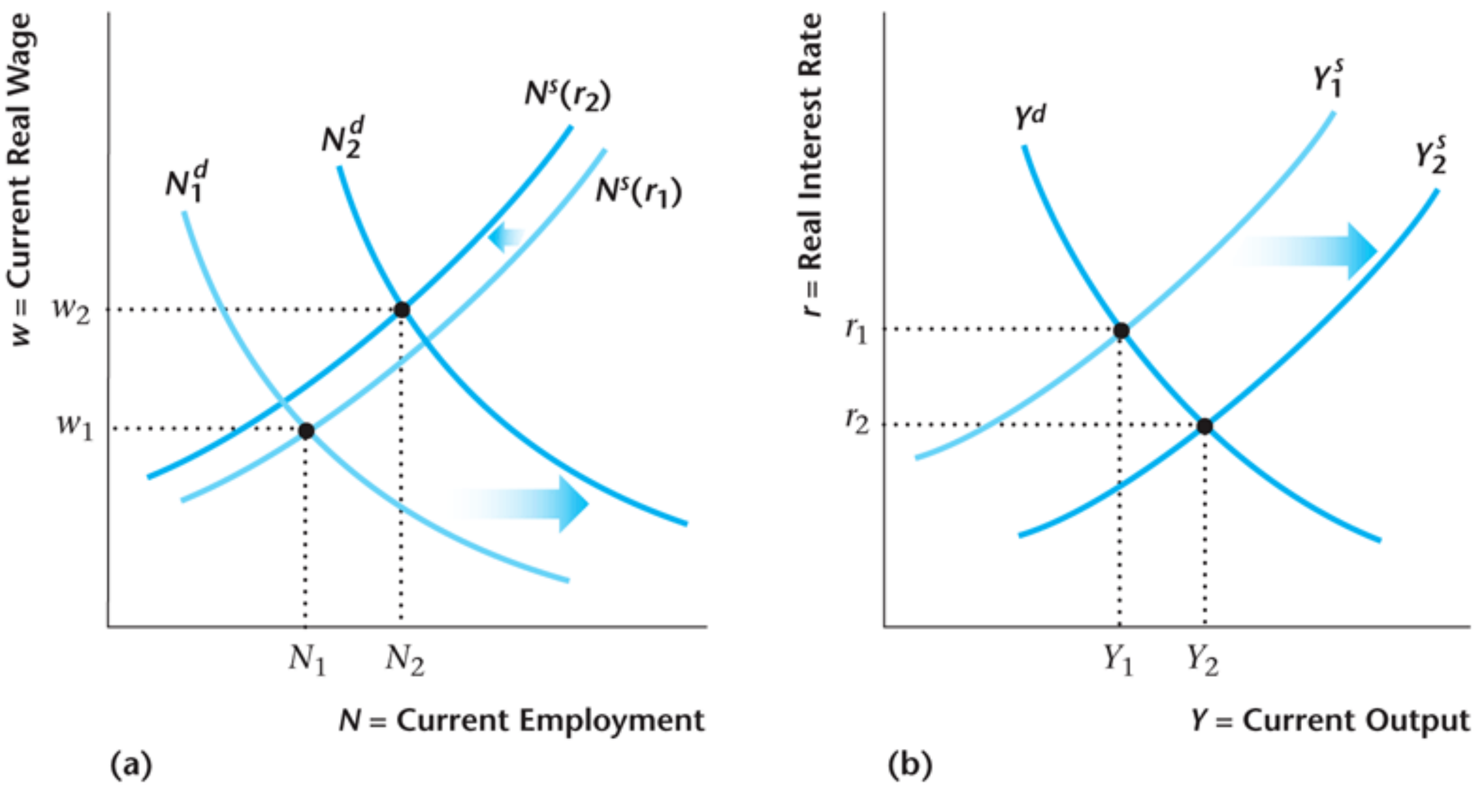
Figure 10.24
The Equilibrium Effects of a Decrease in the Current Capital Stock



Current Total Factor Productivity Increases

- Real interest rate falls, consumption and investment rise, employment rises, real wage rises.
- Productivity shocks are a potential explanation for business cycle.

Figure 10.25
The Equilibrium Effects of an Increase in Current TFP



TFP Expected to Increase in Future – A News Shock

- Output demand curve shifts right.
- Real interest rate rises, investment increases, consumption may rise or fall, employment rises, real wage falls, output rises.
- Important in explaining investment boom in the 1990s in the United States.

Figure 10.26
The Equilibrium Effects of an Increase in Future TFP

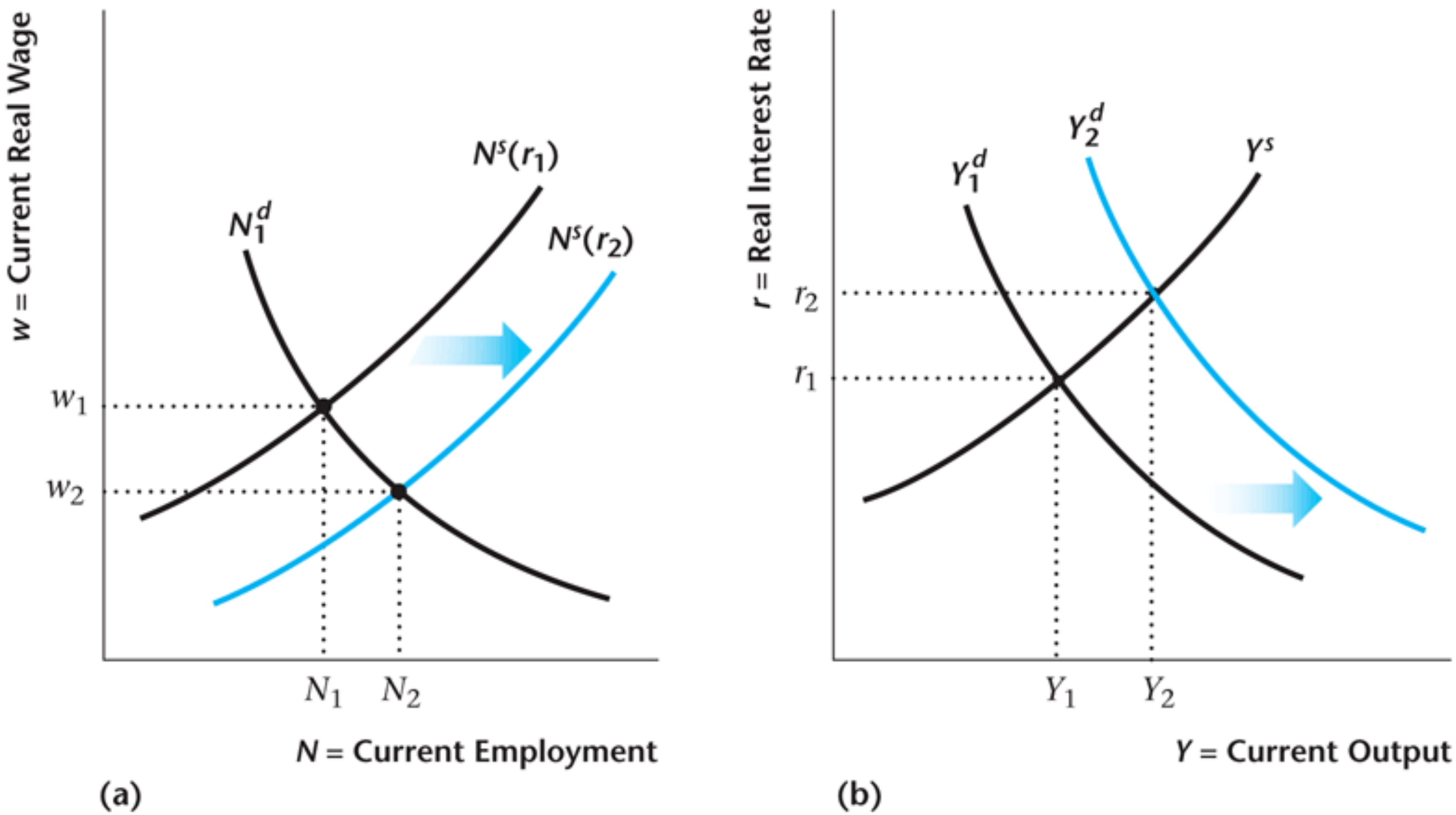
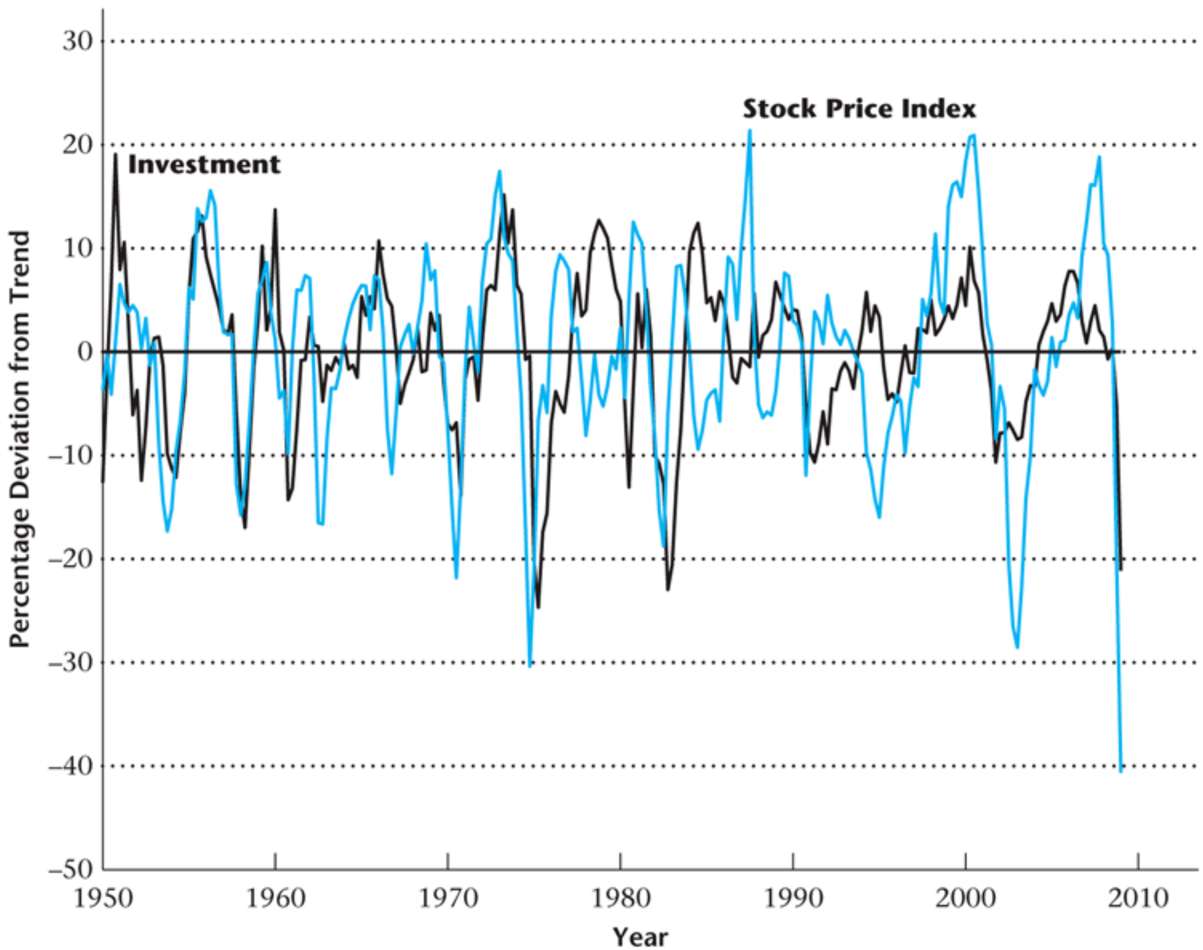


Figure 10.27
% Deviations from Trend in Investment and a Stock Price Index, 1950–2009



An Increase in Credit Market Uncertainty

- Important for the financial crisis.
- Increase in credit market uncertainty affects both consumers and firms – consumers who borrow to finance consumption and firms borrowing to finance investment.
- Increase in interest rate spreads – default premia.

Figure 10.28
The Effect of an Increase in Credit Market Risk

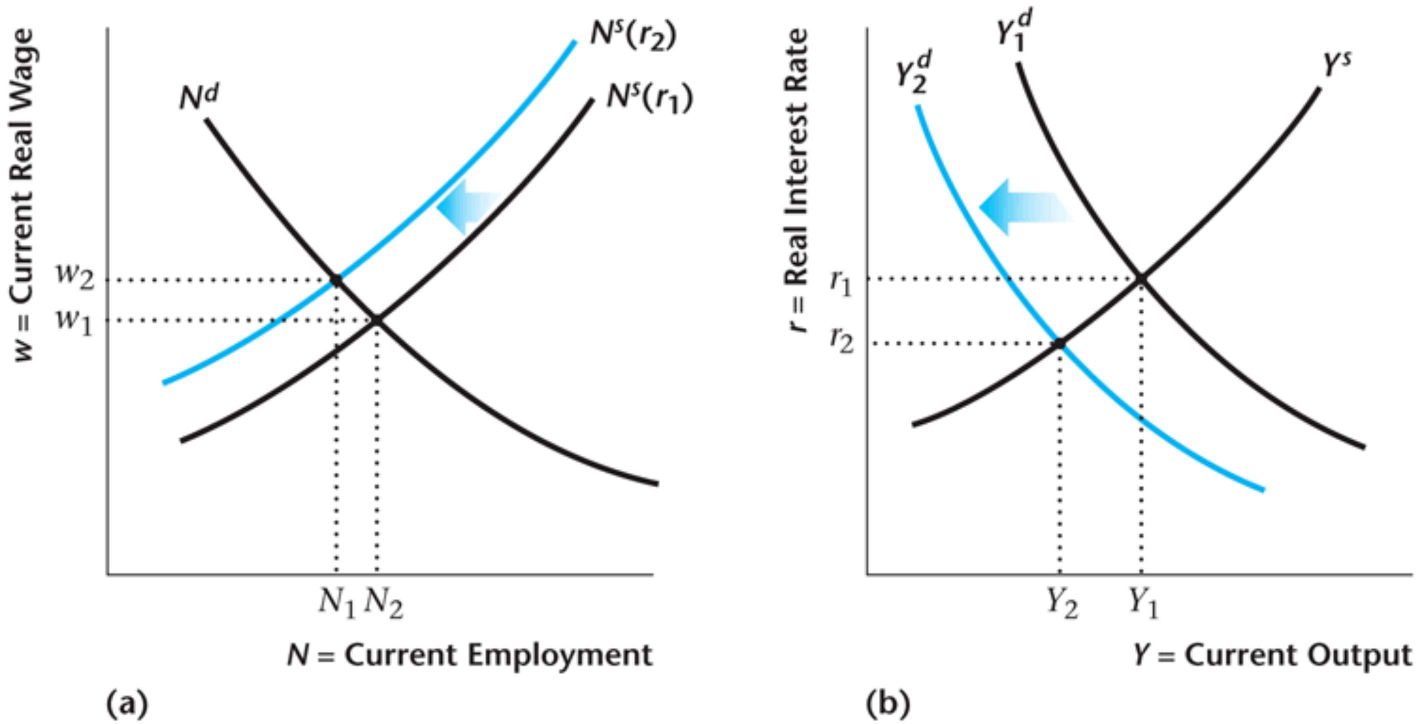


Figure 10.29
Real GDP and the Interest Rate Spread